📘 Dr. Po-Shen Loh's Method for Quadratic Equations

The Goal: To find the roots (solutions) of a quadratic equation mathematically, completely eliminating the need to "guess and check" factors. 🎯 🚫

🔑 The Core Concept & Symbols

Start with any quadratic equation in the standard simplified form:

$$x^2 + Bx + C = 0$$

(Note: If there is a number in front of the x^2, divide the entire equation by that number first so the leading coefficient is 1!) ☝️

- B = The coefficient of the middle x term.
- C = The constant term at the end.
- m = The **average** (midpoint) of the two roots.
- u = The unknown **distance** from the average to the actual roots.

🧠 The Logic (The "Why")

This method relies on two absolute rules of quadratic math:

1. **The Sum Rule:** The sum of the two roots is always equal to $-B$.
    - *Because the sum is $-B$, the **average** (m) of the two roots is exactly half of that: $-\frac{B}{2}$.* ⚖️
2. **The Symmetry Rule:** Because the roots center around their average, we can always define the two roots as:
    - Root 1: $\left(-\frac{B}{2} - u\right)$
    - Root 2: $\left(-\frac{B}{2} + u\right)$
3. **The Product Rule:** The product of the two roots is always equal to C. ✖️

⚙️ The Master Formula (The "How")

By multiplying the two roots together and setting them equal to C, we get:

$$\left(-\frac{B}{2} - u\right)\left(-\frac{B}{2} + u\right) = C$$

Using the algebraic difference of squares rule $(a-b)(a+b)=a^2-b^2$, this simplifies into the only equation you need to solve:

$$\left(-\frac{B}{2}\right)^2-u^2=C$$

🚀 Step-by-Step Example

Let's solve the equation: $x^2-8x+12=0$

Step 1: Identify B and C 🔍

- $B=-8$
- $C=12$

Step 2: Find the average of the roots ($-\frac{B}{2}$) ⚖️

- The sum of the roots is the opposite of B, which is 8.
- The average of the roots is $\frac{8}{2}=4$.
- *This means our roots are $(4-u)$ and $(4+u)$.*

Step 3: Set up the product equation ✖️

Multiply the two roots and set them equal to C (which is 12):

$$(4-u)(4+u)=12$$

Step 4: Solve for u^2 using the difference of squares 📐

$$4^2-u^2=12$$

$$16-u^2=12$$

$$16-12=u^2$$

$$4=u^2$$

Step 5: Find u 🧮

Take the square root of both sides.

$$u=2$$

(Note: We only need the positive 2 because u represents a distance!)

Step 6: Calculate the final roots! 🎉

Plug u back into our root expressions from Step 2:

- **Root 1:** $4 - u \Longrightarrow 4 - 2 = \mathbf{2}$
- **Root 2:** $4 + u \Longrightarrow 4 + 2 = \mathbf{6}$

Final Answer:

The roots are 2 and 6.

If you need to write it in factored form, it is: $(x - 2)(x - 6) = 0$ ✅ ✨